Multimedia Tools and Applications
https://doi.org/10.1007/s11042-024-18733-8

Deep feature-support vector machine based hybrid model for multi-crop leaf disease identification in Corn, Rice, and Wheat

Amit Bhola[1] · **Prabhat Kumar**[1]

Received: 4 August 2023 / Revised: 29 January 2024 / Accepted: 24 February 2024

Abstract
Corn, Rice, and Wheat serve as primary staple foods globally, playing a pivotal role in the economies of numerous countries. Despite their paramount importance, these cereal crops face susceptibility to various diseases, posing a significant threat to production. Timely and accurate disease diagnosis is imperative to mitigate these risks. In response, a range of Machine Learning (ML) and Deep Learning (DL) models have been devised to automate the identification of diseases impacting cereal crops. While DL excels with abundant data, traditional ML may outperform DL with limited data. This study proposes a hybrid model integrating deep transfer learning and ML techniques for identifying crop diseases. The model utilizes pre-trained DenseNet201 weights to extract features, which are subsequently fed into a Support Vector Machine (SVM) for identification. Experimental results show that the hybrid model outperforms several benchmark counterparts. The DenseNet201 deep feature combined with SVM exhibits superior performance, achieving an impressive accuracy score of 87.23% with just 20.2M parameters. The model underwent comprehensive performance assessment using precision, recall, F1 score, and overall accuracy metrics. It effectively classifies healthy and diseased categories for each crop, achieving accuracies of 99.82%, 98.75%, and 84.15% for Corn, Wheat, and Rice, respectively. Moreover, the model's remarkable performance and lightweight nature make it a practical and efficient real-time crop disease detection solution, emphasizing its relevance and significance for farmers and researchers in the agricultural community.

Keywords Plant disease · Transfer learning · Cereal crops · Machine learning · Computer vision

✉ Amit Bhola
amitb.phd19.cs@nitp.ac.in

Prabhat Kumar
prabhat@nitp.ac.in
https://www.nitp.ac.in/profile/?id=prabhat@nitp.ac.in

[1] Department of Computer Science and Engineering, National Institute of Technology Patna, Bihar, India

1 Introduction

Agriculture stands as the foundation of our society, providing sustenance and nourishment to billions worldwide. With the ever-increasing global population, ensuring a stable and secure food supply is paramount [1]. In the fiscal year 2022-2023, agriculture's significant contribution of 18.3% to India's GDP underscored its role as a critical sector [2]. Agriculture fosters economic development and plays a crucial role in generating employment opportunities, supporting the livelihoods of millions across the country [3]. However, ensuring the sustenance of this pivotal sector requires addressing the challenges posed by plant diseases, which can adversely affect crop yields and food production.

An estimated 40% of the world's crop production is lost to disease yearly. [4]. Effective plant disease identification is central to mitigating these challenges, enabling timely intervention and precise management strategies. In addition, early and precise identification and treatment of plant diseases are vital in enhancing agricultural productivity. The leaves of an infected plant are the first to show signs of the disease; hence, they are often utilized to detect and classify disease. An early-stage identification of plant disease and restriction from further spreading is very challenging by optical observation. Traditionally, farmers have used manual inspection to identify plant diseases. However, this method could be more laborious and vulnerable to human inaccuracies. Adopting cutting-edge technologies, such as Internet of Things (IoT) [5, 6] Machine learning [7, 8] and Deep learning [9, 10] has revolutionized detecting and managing crop diseases. These advanced tools enable precise and timely identification of various plant ailments, helping farmers and agronomists implement targeted interventions and optimize disease management strategies.

Research concerning crop disease detection using image data can be divided into two main categories: The first approach involves directly employing deep learning models utilizing plant image data for disease detection. While these models tend to achieve higher accuracy, they require a large dataset for optimal training, and a small dataset may lead to model overfitting and decreased testing accuracy. Additionally, the lack of comprehensive datasets for many plant diseases poses a challenge in this category [11]. The second category involves working with extracted features from image datasets for disease classification using conventional machine learning models like SVM, Decision Tree (DT), Random Forest (RF), Naive Bayes (NB), Ensemble Tree, and K-Nearest Neighbor (KNN). Various statistical tools are needed to extract features from plant image data, such as color, texture, and shape. These features are then used to train a classifier to predict the type of disease present in plants. However, relying solely on manual feature extraction can be an uphill task with lower accuracy. Moreover, applying statistical image processing tools in real-time settings can be challenging.

Integrating both approaches can enhance farmer's effectiveness in disease identification. Deep learning models excel at extracting intricate features and learning task-specific patterns from images, while machine learning models are straightforward, memory-efficient, and exhibit efficient performance. Moreover, these models are easily deployable in practical applications. Therefore, this study suggests a hybrid approach: feature extraction using a benchmark deep learning model and classification using machine learning. Automating plant disease diagnosis through this combined approach is pivotal for significantly improving agricultural productivity with high precision on a broad scale.

Furthermore, this study primarily concentrates on three of the world's most crucial food crops, Rice, Wheat, and Corn (Maize), which collectively contribute to 60% of the dietary energy consumed by humans [12]. The initial step involves feature extraction through a deep-learning architecture. Subsequently, machine learning models classify the disease based on the features obtained from the deep learning model. Finally, the study compares and analyzes the performance and results of disease classification. The noteworthy contributions of the proposed approach for plant disease classification include:

1. The proposed methodology suggests a hybrid approach for disease classification in Corn, Rice, and Wheat. The model combines deep learning (DenseNet201) for extracting important features and feeding these features into machine learning (SVM) for disease classification.
2. Extensive experiments are conducted to demonstrate the efficacy of the proposed approach.
3. Nine benchmark Convolution Neural Network (CNN) models, including MobileNetV2, ResNet50V2, ResNet152V2, DenseNet169, DenseNet201, VGG16, VGG19, InceptionV3, and Xception, are assessed for feature extraction. Additionally, six ML models, DT, KNN, NB, SVM, RF, and XGBoost, are employed for disease classification.

The paper is structured as follows: Section 2 reviews relevant literature. Section 3 describes the proposed approach in detail. The experimental setup is discussed in Section 4, while Section 5 presents the results and related discussion. Finally, Section 6 concludes the work, offering insights into future research opportunities.

2 Related works

In agriculture, plant diseases substantially impact crop yield, resulting in significant economic losses. Researchers have increasingly adopted ML and DL models for disease detection using image data. The literature documents numerous techniques aimed at automating plant disease classification, broadly falling into two categories: (a) ML techniques that depend on handcrafted features and (b) DL techniques. Tables 1 and 2 offer a comparative summary

Table 1 ML-based related works

Reference	Year	Model/Feature(s)	Dataset	Accuracy	Limitations
Khan et al. [13]	2022	RF (HOG, PHOG, GLCM)	Tomato	93.12%	Crop-specific
Kaur et al. [14]	2018	SVM (Color, Texture)	Soyabean	90%	Crop-specific
Basavaiah et al. [15]	2020	DT, RF (Color, Hu Moments, LBP)	Tomato	94%	Crop-specific
Kalyoncu et al. [16]	2016	Linear Discriminant (Color, Texture, Shape)	ICL, Flavia, Swedish	86.8%, 98.6%, 99.5%	Evaluated on Specific Models
Zamani et al. [17]	2022	RBF-SVM (GLCM)	Rice	96%	Crop-specific, Segmentation sensitivity
Rahman et al. [18]	2023	SVM (GLCM)	Tomato	90%	Crop-specific
Ahmed et al. [19]	2023	Ensemble (GLCM)	Self-gathered	90.96%	Limited generalization
Prabhu et al. [20]	2023	BSVM-AOA (GLCM)	PlantVillage	98.6%	Limited to Binary Classification

Table 2 DL-based related works

Reference	Year	CNN model	Dataset	Accuracy	Limitations
Chen et al. [21]	2022	DFCANet	Corn	98.47%	Crop-specific
Genaev et al. [22]	2020	DenseNet201	Wheat	98.00%	Model-specific interpretability, Crop-specific
Faisal et al. [23]	2023	DFNet (MobileNetV2 + NASNetMobile)	Corn Coffee	97.53% 94.65%	Crop-specific test, Multiple extractors
Chen et al. [24]	2022	MobileNet M-Inception	Rice PlantVillage	97.89% 99.21%	Interpretability challenge, Resource-intensive
Wang et al. [25]	2021	ResNeXt-101 VGG16 InceptionV3	PlantDoc PlantVillage	99.7% 75.58%	Low accuracy, Resource-intensive
Liu et al. [9]	2022	PiTLiD (Inception-V3)	Apple	99.45%	Crop-specific, Small datasets challenge
Zhu et al. [26]	2022	LAD-Net	Apple	98.58%	Crop-specific
Ganatra et al. [27]	2020	ResNet50 ResNet101 InceptionV4 VGG16	PlantVillage	99.73%	Resource-intensive, Interpretability challenge
Paymode et al. [28]	2022	VGG16	Grapes Tomatoes	98.40 95.71%	138.4M params, Resource-intensive
Zhou et al. [29]	2021	RDN (Residual nets + Dense nets)	Tomatoes	95.00%	Complex model, Resource-intensive, Crop-specific
Verma et al. [30]	2023	CNN	Cereal Crops	84.4%	Low accuracy, Complex model
Haridasan et al. [31]	2023	CNN	Rice	91.45%	Limited Disease Variety

of the critical aspects of previous works in ML-based and DL-based disease identification, respectively.

2.1 Machine learning approaches

Recognition of leaf diseases using machine learning methods primarily depends on manually engineered features that typically involve a three-step procedure: (a) First, plant disease images are segmented using established techniques like color conversion, edge-based segmentation, and mathematical morphology. (b) Next, attributes are derived from the segmented images, encompassing texture, shape, size, and color information. (c) Finally, ML models classify plant diseases based on the extracted features.

Khan et al. [13] suggest an approach for identifying diseases in tomato plants by integrating color balancing and superpixel techniques. The methodology transforms input images into color-balanced representations and applies superpixel operations for effective segmentation, overcoming challenges like uneven illumination and complex backgrounds. Employing features like the Gray Level Co-occurrence Matrix (GLCM), Pyramid of HOG (PHOG), and Histogram of Gradients (HOG), the system identifies the disease-infected parts in the image. Random Forest (RF) is chosen as the classifier, and the method demonstrates robustness with a 93.12% accuracy on diverse datasets.

The research by Basavaiah et al. [15] involved extracting and combining diverse features such as color histograms, Haralick texture, Hu moments, and local binary pattern (LBP). These merged features were input for decision trees and random forest classifiers to categorize leaf diseases. The outcomes revealed a noteworthy accuracy of 94% for the random forest classifier.

Kalyoncu et al. [16] combine multiple descriptors, including extracting color, shape, geometric, and texture features from leaf images. Introducing a novel variant of the local binary pattern called sorted uniform LBP enhances texture description. After extracting features, the subsequent step involves classification using the linear discriminant classifier. The approach achieves high accuracy scores of 86.8%, 98.6%, and 99.5% on the ICL, Flavia, and Swedish datasets. Kaur et al. [14] proposed a method to classify soybean diseases based on color and texture features, employing a support vector machine and obtaining an accuracy of 90.00%.

Zamani et al. [17] introduced an ML model for Rice disease detection. n. They utilized principal component analysis (PCA) for feature extraction and employed algorithms including SVM, RF, ID3, and RBF-SVM for disease classification. The best-performing model was RBF-SVM, achieving an accuracy of 96.00%.

Rahman et al. [18] suggested a system for disease identification in tomato crops. It utilizes the GLCM algorithm to extract statistical features, which are then fed into SVM for disease classification. The system achieves accuracies of 100%, 95%, 90%, and 85% for healthy, early blight, septoria leaf spot, and late blight disease, respectively.

Ahmed et al. [19] proposed ML-based disease detection models such as RF, linear regression, NB, neural networks, and SVM. The models are assessed using classification metrics, achieving an accuracy of 90.96% with the ensemble model.

Prabhu et al. [20] proposed the Boosted Support Vector Machine-based Arithmetic Optimization Algorithm (BSVM-AOA) for plant disease identification. The method employs image segmentation using feature extraction through the GLCM to address low precision and high calculation costs, achieving an accuracy of 98.6%.

These studies illustrate continuous research in ML for plant disease classification, highlighting diverse benchmark models that enhance accuracy and tackle challenges such as limited training data. The subsequent subsection delves into various deep learning techniques.

2.2 Deep learning approaches

Recently, researchers have directed their attention toward employing DL techniques for plant disease classification [32]. DL, a subset of ML, excels at learning and recognizing patterns from data, making it highly suitable for plant disease identification and distinguishing between healthy and diseased plants.

Chen et al. [21] propose DFCANet, a CNN designed for corn disease detection. Trained on a dataset comprising 1,200 corn leaf images with various diseases such as leaf rust, leaf spot, and smut, the CNN achieves an impressive accuracy of 98.47%.

Faisal et al. [23] introduced DFNet, a CNN model incorporating MobileNetV2 and NAS-NetMobile for disease classification. DFNet attains an accuracy of 97.53% for corn and 94.65% for coffee plants. By leveraging diverse features extracted from multiple CNNs, DFNet demonstrates superior and consistent performance, contributing to timely crop production preservation.

Chen et al. [24] proposed the MobInc-Net model for crop disease identification. The method employs a modified Inception module (M-Inception) in conjunction with MobileNet, yielding high-quality features, achieving an accuracy of 99.21% and 97.89% on two different datasets [33].

Wang et al. [25] proposed Trilinear Convolution Neural Networks (ResNeXt101, VGG16, and InceptionV3) as the base networks for disease classification on datasets [33, 34] and attained an accuracy of 99.7% and 75.58%, respectively. Liu et al. [9] proposed PiTLiD based on InceptionV3 for disease identification. The suggested approach attains an accuracy of 99.45% on a dataset collected from PlantVillage [33].

Zhu et al. [26] proposed a DL model for the early detection of pests and diseases to obtain an accuracy of 98.58% affecting apple leaves. Paymode et al. [28] fine-tuned VGG16 architecture for identifying plant diseases on manual data collected and achieved an accuracy of 98.40% and 95.71% for two different plant diseases. Saleem et al. [35] used the Xception architecture with Adam Optimizer to increase accuracy to 99.81%. Ganatra et al. [27] used VGG16, InceptionV4, ResNet101, and ResNet50 for plant disease identification. The results indicated that ResNet101 surpassed the other models, achieving an accuracy of 99.73%. In another technique, Zhou et al. [29] designed a residual dense network with a 95% accuracy for disease identification.

Singh et al. [36] proposed a DL model to identify diseased mango plants comprising 1070 images, and the model attains an accuracy of 97.13%. Verma et al. [30] proposed a DL model for disease classification in cereal crops. The model attains 84.4% accuracy in classifying various healthy and infected categories for the major cereal crops. Haridasan et al. [31] introduce a DL model for disease classification in paddy plants. The automated approach employs machine and deep learning models to identify diseases. Integration of the SVM classifier and CNN facilitates disease identification, achieving an accuracy of 91.45%.

These works demonstrate ongoing research and advancements in DL for plant disease identification, showcasing various pre-trained models and techniques to improve disease classification accuracy. Further, existing architectures like DenseNet, VGG, Xception, and ResNet exhibit better performance in crop image classification. Nonetheless, these networks encounter challenges related to convergence, overfitting, and vanishing gradient problems. DenseNet-based CNN provides precise options for plant disease classification to address these issues, offering enhanced feature extraction capabilities.

The research by Akkem et al. [37] offers a comprehensive review of smart farming using Artificial Intelligence (AI) techniques, focusing on ML, DL, and time series analysis. The paper also explores various ML methods such as SVM, KNN, Fuzzy networks, ARIMA, DT, and Ensemble learning. It emphasizes the integration of multiple ML methods and the combination of ML and DL for improved results. Additionally, the study by Orchi et al. [38] found that SVM, Artificial Neural Network (ANN), CNN, and Recurrent Neural Network (RNN) are advantageous for plant disease detection. ML algorithms, including SVMs, are relatively simple to train and can achieve high accuracy. ANN training is more complex than SVMs but can be sensitive to overfitting for small training datasets.

The choice of which algorithm to use for plant disease identification depends on the unique characteristics of the disease. For instance, when dealing with different scenarios in image analysis, selecting the appropriate algorithm depends on the nature of the problem. For example, a CNN is suitable for identifying plant diseases in images. However, ML methods could be more optimal if the disease involves internal damage and limited training data. On the other hand, if the disease exhibits changes over time in a sequential manner, an RNN is the most appropriate choice. Further, continuous monitoring is required to verify that models continue to provide accurate outputs, especially when there are changes in the underlying conditions [39]. The models available so far were developed and documented to detect diseases in individual crops such as Corn, Rice, or Wheat. This study proposes a joint model based on DL and ML that can accurately predict Corn, Rice, and Wheat diseases.

3 Methodology

This section emphasizes the methodology used in this work by highlighting model selection, outlining the requirements, and discussing the choice between machine learning, deep learning, or a combination of both. Further, the proposed hybrid model and its architecture are elaborated in the end.

3.1 Model selection

Model selection involves choosing the most suitable model from a set of candidates based on performance criteria to effectively address the target problem (such as plant disease identification). This selection process can be applied across various types of ML or DL models (including RF, SVM, KNN, ANN, CNN) and even among models of the same type with different configurations, such as distinct hyperparameters (for instance, different kernels in an SVM, varying ANN neurons or layers). Additionally, the task may include designing experiments to ensure the collected data is well-suited for the model selection problem. The simplest, memory-efficient, and real-time adaptability model is often the best choice when faced with candidate models with similar predictive or explanatory power.

3.1.1 Machine learning models

Machine learning, mainly supervised learning models, has been instrumental in advancing plant disease identification tasks. Classification models such as Support Vector Machines, Random Forests, Decision Trees, Naive Bayes, and K-Nearest Neighbors have effectively distinguished healthy and diseased plants. Tree-based algorithms, including Random Forest and Extreme Gradient Boosting, leverage the strengths of decision trees while addressing their limitations. Their ability to handle large datasets and provide robust predictions makes them valuable assets. SVM, known for its ability to find optimal hyperplanes, proves effective in scenarios where classes are linearly separable. SVM also excels in handling complex decision boundaries and capturing nonlinear relationships, often characteristic of plant disease patterns.

Further, Naive Bayes, known for its simplicity and efficiency, operates on conditional probability. It assumes that features are conditionally independent, making it particularly effective for tasks where this assumption holds. On the other hand, K-Nearest Neighbors, a versatile and intuitive algorithm, classifies samples based on the majority class among their K-nearest neighbors. In this work, the mentioned machine learning models are explored for classifying plant disease, and SVM outperformed.

SVM stands out in its ability to handle high-dimensional data effectively and discern intricate patterns within image datasets. This capability is further enhanced by its flexibility in utilizing various kernel functions, allowing it to capture the complex relationships inherent in the data. Leveraging the kernel trick, SVM beats the limitations of linear decision boundaries by transforming the input space into a higher dimension. Notably, SVM's efficiency is exemplified in its utilization of a subset of training points, termed support vectors, making it a memory-efficient choice less prone to overfitting. Particularly advantageous in scenarios with approximately linearly separable data, SVM excels in determining an optimal hyperplane that maximizes the margin between classes, thereby establishing a well-defined decision boundary.

3.1.2 Deep learning based pre-trained models

Traditionally, deep learning involves incorporating multiple layers of nonlinear processing modules to extract features. These layers are interconnected, each taking the previous layer's output as input. Researchers have tried to make deep learning networks deep and large to explore their capabilities. However, one major challenge encountered is the issue of exploding or vanishing gradients, which can hinder training. Consequently, researchers have proposed various transfer learning-based pre-trained CNN architectures for deep transfer learning to address this challenge.

Transfer learning is a powerful paradigm that leverages pre-trained neural network architecture to enhance the performance of a model on a new, related task. In this approach, a neural network is initially trained on a large dataset and learns to extract general features and representations. These learned features are transferred and fine-tuned on a smaller dataset specific to the target task. Transfer learning is particularly advantageous when working with limited data for a new task, as it allows the model to capitalize on the knowledge acquired from the source task.

In this work, various pre-trained models, such as VGG16, VGG19, ResNet50V2, ResNet152V2, InceptionV3, DenseNet169, DenseNet201, MobileNetV2, and Xception, are explored for feature extraction in cereal plant disease classification. Table 3 presents an overview of these pre-trained deep transfer learning models, highlighting their size, parameters, depth, and execution time in milliseconds (ms). Among these models, MobileNetV2, DenseNet169, and DenseNet201 stand out for having the smallest size and parameters.

Table 3 Pre-trained models comparison

Pre-trained Models	Size(MB)	Parameters(M)	Depth	Time (ms) per Inference Step (GPU)
VGG16	528	138.4	16	4.2
VGG19	549	143.7	19	4.4
ResNet50V2	98	25.6	103	4.4
ResNet152V2	232	60.4	307	6.6
DenseNet169	57	14.3	338	6.3
DenseNet201	80	20.2	402	6.7
InceptionV3	92	23.9	189	6.9
MobileNetV2	14	3.5	105	3.8
Xception	88	22.9	81	8.1

DenseNet is parameter-efficient due to its dense block architecture, where each layer receives feature maps from all preceding layers. This design reduces the number of parameters compared to traditional architectures, making it well-suited for transfer learning on limited datasets, such as those in plant disease identification. DenseNet201 is an extension of DenseNet with a deeper architecture, featuring 201 layers that involve extensive feature fusion through dense connections. This depth allows it to capture intricate patterns and abstract features, presenting diverse representations crucial for distinguishing subtle variations in plant diseases. Remarkably, despite its depth, DenseNet201 maintains a relatively lightweight design in transfer learning. This characteristic translates into low computational overhead during training and inference, striking an optimal balance between performance and computational cost.

3.2 The proposed hybrid model

In agriculture, acquiring a large labeled dataset for training machine learning models is challenging, leading to concerns about overfitting, especially with smaller training samples. Plant disease identification tasks often involve image datasets with a relatively high number of features due to high pixel values. Also, disease detection often involves intricate and complex patterns that simpler machine learning models may not capture. Deep learning models have recently excelled in such tasks, but they can be memory-intensive and require extensive training datasets. Also, the performance of traditional ML models often relies on well-engineered features. Generating meaningful features necessitates a deep understanding of the domain, and the quality of these features can constrain the model's performance. Hence, combining deep learning and traditional machine learning techniques, a hybrid approach enhances plant disease identification systems' robustness, interpretability, and effectiveness. Deep learning models can automatically extract complex features, which can be used as input for traditional machine learning models, enhancing the overall discriminative power.

Figure 1 outlines the proposed architecture, encompassing two pivotal modules for disease identification. The initial module serves as a feature extractor, employing deep transfer learning techniques, while the second module employs classical machine learning for disease classification. Deep features are derived from the fully connected layer and are input into

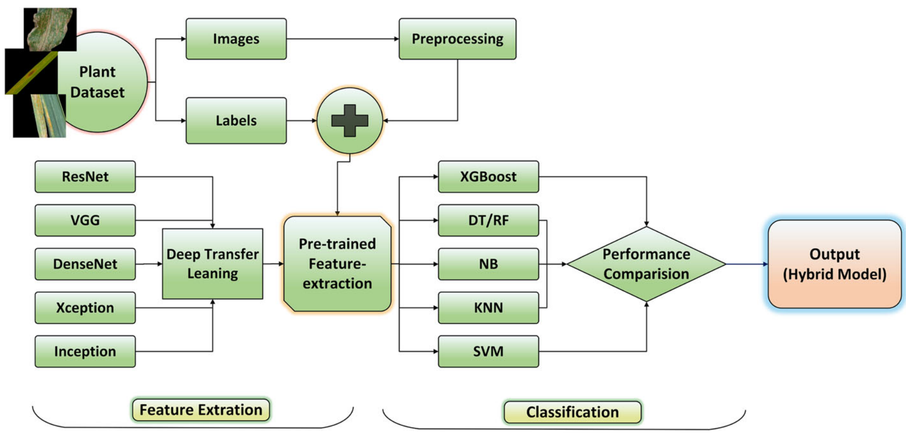

Fig. 1 Proposed architecture

the classifier for training. The inherited layers from the pre-trained models remain frozen, ensuring they are untrainable. The ML classifier utilizes these extracted deep features for classification. This hybrid model strategically combines the strengths of pre-trained DL models for feature extraction with ML, offering high speed and less overfitting and avoiding prolonged training processes while achieving enhanced classification accuracy. The proposed hybrid model involves the following steps:

1. Gathering images of both diseased and healthy cereal plant leaves.
2. Pre-processing data involving resizing images to $224 \times 224 \times 3$ dimensions, employing augmentation techniques for size adjustment.
3. Extraction of features from fully connected layers of the pre-trained network.
4. Classification utilizing deep features with a machine learning classifier, followed by performance measurement.

The process of choosing the optimal machine learning and pre-trained models for disease identification included evaluating various options. Due to its outstanding performance, DenseNet201 was identified as the optimal feature extractor for crop disease identification. Additionally, SVM was chosen for classification based on experimental results, affirming its status as one of the best-performing models for classifying crop diseases, with added benefits, as mentioned earlier.

The proposed hybrid model (DenseNet201+SVM) is depicted in Fig. 2. DenseNet201 boasts compact dimensions, with just 80MB and 20.0M parameters, rendering it smaller than most other pre-trained models (excluding DenseNet169 and MobileNetV2). This compactness makes it a feasible choice for real-time scenarios. Further, SVM supports linear and nonlinear solutions using kernel tricks and better handles outliers. Hence, to enhance the model's performance and prevent overfitting during the classification process, the final layer of DenseNet201 was eliminated and substituted with a Support Vector Machine. This modification aims to improve the overall performance of the model.

4 Experiment

The following section outlines the steps involved in image augmentation, provides information on how the procedure was implemented, discusses the dataset used, explains the evaluation metrics employed, and presents the training specifications utilized for the disease identification of three cereal crops in this study.

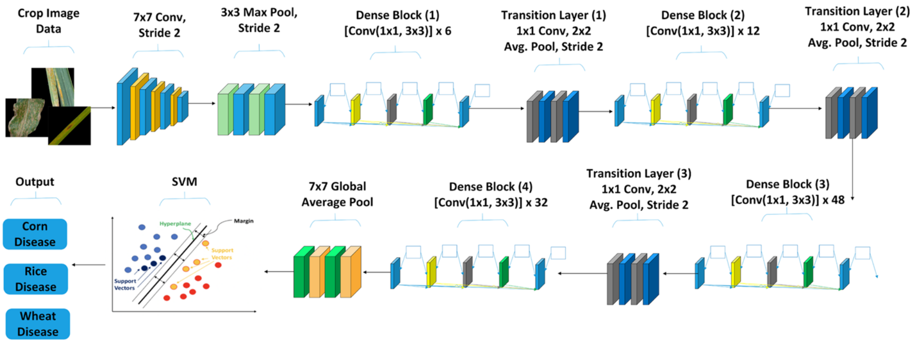

Fig. 2 Proposed hybrid model (DenseNet201+SVM)

4.1 Image augmentation

The images in a dataset can be of different sizes. However, to accommodate the requirements of transfer learning models, it is necessary to ensure they are of the same input size. Hence, all images are resized to dimensions of 224x224. Additionally, rescaling is applied to normalize the pixel values of the images, ensuring they fall within the range of [0, 1].

4.2 Implementation set-up

For this research work, the experimentation was conducted on Google Colab. Python was employed to develop the proposed model, primarily utilizing libraries such as Pandas, Keras, Numpy, and Tensorflow.

4.3 Dataset

The dataset used in this study is compiled from diverse sources focusing on plant diseases in Corn, Rice, and Wheat crops. Specifically, images of Corn are derived from the PlantVillage Dataset [33], Rice diseases data is acquired from Kaggle [40], and the Wheat diseases dataset is aggregated from two separate repositories on Kaggle [41, 42]. Figure 3 depicts sample images showcasing the data for three cereal crop categories.

The resulting dataset is divided into a train, test, and validation set, with a split of 70:15:15. Table 4 displays the overall distribution of various diseases within the crops. The generated dataset exhibits an imbalanced nature, where Healthy Rice and Wheat stem rust comprise 1192 and 1188 images, respectively, while Corn grey spot disease comprises only 408 images.

4.4 Performance metrics

This study encompasses 12 distinct diseased and healthy categories associated with Corn, Rice, and Wheat crops. Consequently, a binary and multi-class classification approach is employed, and a classification report is generated to compute the performance of ML models.

- **True Positive** (TP): Correctly identified instances of positive cases.
- **True Negative** (TN): Correctly identified instances of negative cases.
- **False Positive** (FP): Incorrectly identified instances as positive when they are negative.
- **False Negative** (FN): Incorrectly identified instances as negative when they are positive.

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{4}$$

Equation (1) represents accuracy, calculating the ratio of correct predictions by the total number of predictions made by the model. Equation (2) defines precision, which measures

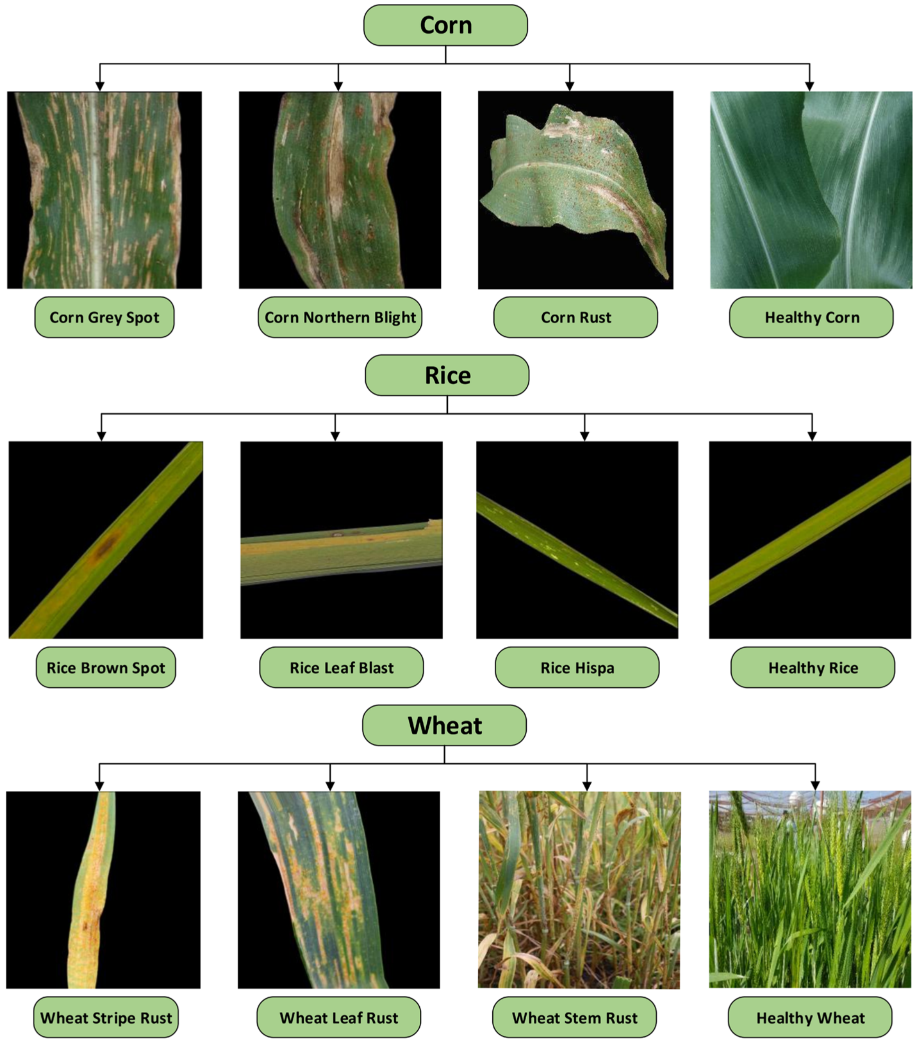

Fig. 3 Sample crop image dataset

the model's precision level. It indicates that the proportion of correctly predicted cases was positive. In contrast, (3) represents recall, also known as sensitivity or true positive rate, and measures the proportion of actual positive cases the model correctly identifies. The F1 score, as calculated in (4), represents the harmonic mean of precision and recall.

5 Results

In this section, the effectiveness of the proposed model has been rigorously evaluated through comparative experiments. The results of these experiments are comprehensively analyzed, comparing them with state-of-the-art approaches. Additionally, the adaptability of

Table 4 Crop disease dataset distribution

Crop	Class	Class No. (Classification)			Image Data			Total & Class
		Binary	Multi-Class		Train	Test	Validate	Distribution (%)
Corn	Corn Grey Spot		0	0	286	61	61	408 (4%)
	Corn Northern Blight	0	1	1	552	118	118	788 (8%)
	Corn Rust		2	2	668	142	142	952 (10%)
	Healthy Corn	1	3	3	650	139	139	928 (10%)
Rice	Rice Brown Spot		0	4	367	78	78	523 (6%)
	Rice Hispa	0	1	5	397	84	84	565 (6%)
	Rice Leaf Blast		2	6	547	116	116	779 (8%)
	Healthy Rice	1	3	7	836	178	178	1192 (12%)
Wheat	Wheat Leaf Rust		0	8	530	113	113	756 (8%)
	Wheat Stem Rust	0	1	9	832	178	178	1188 (12%)
	Wheat Stripe Rust		2	10	580	124	124	828 (9%)
	Healthy Wheat	1	3	11	494	105	105	704 (7%)
					6739	1436	1436	9611 (100%)

the framework is explored through the lens of a discussion section, providing a holistic view of its performance and potential.

5.1 Disease classification

This section outlines the outcomes of the experiments performed on the suggested model under different scenarios for classifying disease in three cereal crops. These scenarios include (a) Binary classification, distinguishing between healthy and diseased classes for individual crops; (b) Multi-class classification, (i) classifying diseases for individual crops separately; and (ii) classifying healthy and diseased categories of all three crops collectively.

5.1.1 Healthy vs diseased crops

Table 5 presents the binary classification accuracy achieved by the suggested model in distinguishing between healthy and infected cereal crops. The outcomes reveal the topmost accuracy of 99.82% is obtained for Corn, while the lowermost accuracy of 84.15% is obtained for Rice. The reduced accuracy in Rice can be attributed to the challenges in differentiating it from other classes due to similarity of certain disease-related features, leading to misclassification. Additionally, the presented model is assessed for individual crops separately and all crops collectively, and the obtained outcomes are discussed in the subsequent subsections.

Table 5 Binary classification

Crop	Accuracy (%)
Corn	99.82
Rice	84.15
Wheat	98.75

Table 6 Multi-class classification (Corn)

Class	Precision	Recall	F1-score
Corn Grey Spot	92.72	83.96	87.93
Corn Northern Blight	91.93	96.61	94.21
Corn Rust	100.00	100.00	100.00
Healthy Corn	100.00	100.00	100.00
Accuracy			96.95

5.1.2 Crop-specific disease identification

This section delves into the outcomes of the individual disease classification of all three cereal crops by the suggested hybrid model. Table 6 presents a classification report for Corn disease identification. Remarkably, the suggested approach achieves precision and recall of 100% for Corn Rust and Healthy Corn, indicating that all healthy Corn images are accurately classified. Nevertheless, the overall accuracy diminishes to 96.95% because of misclassifying two infected Corn image classes.

The outcomes for disease identification in Rice are outlined in Table 7. The accuracy for Rice crops decreases to 74.03% for multi-class classification, as opposed to the 84.15% achieved for the binary classification. Likewise, the suggested model accurately classifies more than 75% of the images for all four crop classes of Rice. However, it accurately identifies 50.47% of Rice Hispa Disease only. Moreover, with a precision of 60.75%, it can be deduced that the proposed model categorizes approximately 39 out of every 100 images as Rice Hispa disease, corresponding to another category. The misclassification of Rice Hispa could be attributed to the lack of discernible characteristics that differentiate it from other classes.

The results of the suggested model for identifying disease in Wheat crops are displayed in Table 8. The model attains an accuracy of 93.84% for the Wheat crop, representing a decrease of 4.91% compared to binary classification. Additionally, each class achieves a recall of more than 91%, signifying that the proposed model accurately classifies at least 91 out of 100 images in these classes. Moreover, all classes, except Wheat Leaf Rust, achieve a precision of over 93%. Nevertheless, only 89% of the images belonging to Wheat Leaf Rust are accurately classified, with the remaining 11% being misclassified into other classes.

5.1.3 Combined crop disease identification

Table 9 presents the classification report of the proposed model for the combined evaluation of all 12 classes for three crops. Among all the classes, the model achieves the highest accuracy in classifying Corn Rust and Healthy Corn with precision, recall, and F1-Score values

Table 7 Multi-class classification (Rice)

Class	Precision	Recall	F1-score
Rice Brown Spot	77.10	75.38	76.23
Rice Hispa	60.75	50.47	55.03
Rice Leaf Blast	76.01	66.89	71.11
Healthy Rice	76.19	89.21	82.12
Accuracy			74.03

Table 8 Multi-class classification (Wheat)

Class	Precision	Recall	F1-score
Wheat Leaf Rust	89.56	91.15	90.35
Wheat Stem Rust	93.75	92.69	93.22
Wheat Stripe Rust	96.03	97.58	96.80
Healthy Wheat	96.11	94.28	95.19
Accuracy			93.84

of 100%. The result indicates that every prediction for this class was accurate, and none were misclassified. Furthermore, the model correctly classifies Corn Northern Blight and Corn Grey Spot with precision values of above 91%. This result indicates the model's effectiveness in accurately categorizing all Corn images. Nevertheless, some misclassifications were observed, notably in Corn Grey Spot, which achieved a recall of 83.6%.

Further, Rice Hispa achieves the lowest precision value of 50.76%, recall value of 42.28%, and F1-Score of 44.29%, and only 50% of the images are accurately identified as Rice Hispa. The low results for the Rice crop can be attributed to the imbalanced dataset, as depicted in Table 4, where Rice Brown Spot and Rice Hispa account for only 6%, while Healthy Rice accounts for 12%. The result obtained by the model for Wheat correctly classifies all classes with precision values above 91%. Further, the result for Wheat Leaf Rust is lowest with precision, recall, and F1-score of 89.56%, 91.15%, and 90.35%, respectively. The low values could be because the lowest 8% of image data is available for Wheat Leaf Rust. Broadly, the suggested hybrid model attains an accuracy of 87.23%.

5.2 Comparison with other models

To evaluate the performance of the proposed hybrid approach, features were extracted from a range of benchmark pre-trained models, including MobileNetV2, ResNet50V2, ResNet152V2, DenseNet169, DenseNet201, VGG16, VGG19, InceptionV3, and Xception. Subsequently, machine learning models including Decision Tree, K-Nearest Neighbors,

Table 9 Multi-class classification report (Combined Crops)

Classes	Precision	Recall	F1-Score
Corn Grey Spot	92.72	83.6	87.93
Corn Northern Blight	91.93	96.61	94.21
Corn Rust	100.00	100.00	100.00
Healthy Corn	100.00	100.00	100.00
Rice Brown Spot	67.1	65.38	66.23
Rice Hispa	50.76	42.28	44.29
Rice Leaf Blast	66.1	56.89	61.11
Healthy Rice	65.43	79.77	71.89
Wheat Leaf Rust	89.56	91.15	90.35
Wheat Stem Rust	93.75	92.69	93.22
Wheat Stripe Rust	95.96	95.96	95.96
Healthy Wheat	96.11	94.28	95.19
Accuracy			87.23

Table 10 Accuracy of various hybrid (ML+DL) models

ML + DL Model	DenseNet121	DenseNet169	DenseNet201	VGG16	VGG19
DT	52.08	52.78	52.22	48.18	45.96
KNN	71.51	73.25	73.11	62.39	62.53
NB	69.35	72.42	73.46	47.35	47.42
Linear SVM	**83.84**	**84.26**	**87.23**	**80.77**	**79.24**
Poly SVM	80.84	82.72	83.35	73.81	72.98
RBF SVM	82.45	83.84	85.23	70.61	69.7
RF	74.09	75.41	75.97	71.51	70.82
XGBoost	81.85	82.24	83.98	77.36	77.01

The bold entries are the best results acheived by the proposed model as mentioned in Result section

Naive Bayes, Support Vector Machine, Random Forest, and XGBoost were trained and evaluated on the extracted features for plant disease classification. As depicted in Tables 10 and 11, the outcomes underscore the exceptional performance of the proposed DenseNet201+LinearSVM model, securing the highest accuracy at 87.23%. Notably, this impressive accuracy is achieved with a modest parameter count of 20.2 million, emphasizing the model's efficiency in identifying diseases proficiently across the three cereal crops.

The Linear SVM outperformed other machine learning algorithms in combination with the proposed DL model. Conversely, tree-based algorithms, especially Decision Trees, exhibit the poorest performance with an accuracy of 43.31% when combined with InceptionV3, possibly due to challenges in capturing spatial relationships and intricate patterns in image data. Similarly, Naive Bayes and K-Nearest Neighbors show suboptimal performance of 44.49% and 60.16% in combination with InceptionV3 and ResNet152V2, respectively. The struggles of KNN in navigating high-dimensional feature spaces might contribute to its diminished performance.

Moreover, all other deep learning models with SVM performed worse in comparison to DenseNet201, except DenseNet169 and MobileNetV2. The combination of DenseNet169 and MobileNetV2 with Linear SVM achieves accuracies of 84.26% and 79.12%, respectively, marking a notable 2.97% and 8.11% lower accuracy than the proposed model. It is crucial to note that, despite DenseNet169 and MobileNetV2 having fewer parameters than the proposed model, the achieved accuracy underscores the pivotal role of DenseNet201's architecture.

Table 11 Accuracy of various hybrid (ML+DL) models

ML + DL Model	MobileNetV2	ResNet50V2	ResNet152V2	InceptionV3	Xception
DT	45.11	48.32	50.97	43.31	49.86
KNN	63.67	62.81	60.16	70.68	73.39
NB	44.49	48.18	45.75	66.29	61.01
Linear SVM	**79.12**	**83.49**	**82.93**	**81.12**	**84.12**
Poly SVM	71.05	73.88	67.2	80.5	76.94
RBF SVM	71.61	82.17	79.87	80.64	80.29
RF	70.47	75.13	73.39	70.54	74.93
XGBoost	71.22	80.91	79.52	77.78	78.13

The bold entries are the best results acheived by the proposed model as mentioned in Result section

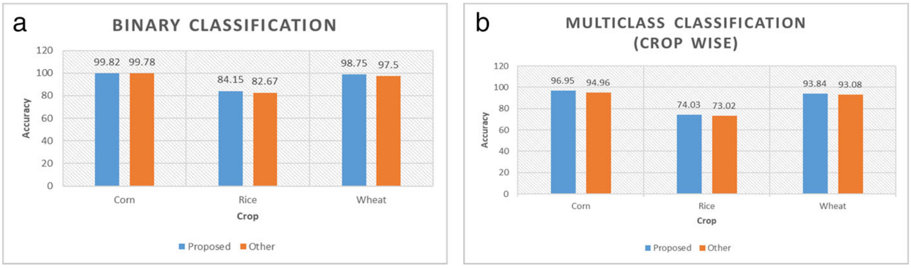

Fig. 4 Comparison with others work (a) Binary classification (b) Multi-class classification

5.3 Comparison with other works

The research findings have been compared with other state-of-the-art cereal leaf disease classification approaches to validate them. Figure 4 provides a comparison demonstrating the superiority of this study over previous works that employed computer vision for classifying cereal crop leaf diseases. Specifically, the results are compared with the work of Verma et al. [30], which proposed a CNN model for disease identification in cereal crops.

Verma et al. [30] achieved an overall accuracy of 84.4% with 387,340 parameters; however, the model size is not provided. In contrast, this approach achieved an accuracy of 87.23% (a notable increase of 2.83%) with 20.2M parameters and a size of 80MB. Despite utilizing more parameters, the proposed hybrid model can effectively be used for various plants even with limited training data. Additionally, the proposed model exhibits remarkable versatility by correctly classifying each crop's healthy and diseased categories, achieving higher accuracy with differences of 0.04% for Corn, 1.48% for Rice, and 1.25% for Wheat in binary classification. Furthermore, for multiclass classification of each crop, the model demonstrates its outperformance with accuracy differences of 1.99% for Corn, 1.01% for Rice, and 0.76% for Wheat. These results demonstrate that this approach surpasses existing state-of-the-art techniques for classifying cereal diseases.

5.4 Discussion

This study proposed a notable hybrid model for disease identification in three major cereal crops, demonstrating its superiority over benchmark models. The proposed model, integrating DenseNet201 for feature extraction and Support Vector Machine, particularly Linear SVM, for classification, presents a compelling solution for multi-crop leaf disease identification in Corn, Rice, and Wheat. The model excels in feature extraction for disease classification using DenseNet201 with a modest set of parameters (20.2 million) and minimal space occupancy (80 MB). It is essential to note that the superiority of a model depends on the nature of the data and the specific characteristics of the problem at hand. Choosing the best model involves experimentation and validation on the specific dataset and problem domain.

Tables 10 and 11 prove that Linear SVM, along with DenseNet201, is the best-performing model among various tested models on the plant disease data. DenseNet201, an extension to DenseNet, involves extensive fusion through dense connections. The added depth contributes to advanced high-level feature extraction, capturing intricate patterns and abstract features crucial for discerning subtle differences in plant diseases. Despite having significantly fewer parameters than VGG19, the model achieves an impressive accuracy of 87.23%, outperforming VGG19 by a notable 5.99%. The model demonstrates an 8.11% accuracy increase while employing few more parameters than MobileNetV2, thus highlighting its efficiency.

In this study, DenseNet201 produces a substantial feature vector for each image, particularly with a feature size that makes the linear kernel suitable for disease classification. Moreover, the choice of SVM for disease classification enhances class separation and reduces expected prediction error, making it the best-performing model among various tested ML models. SVM's proficiency in handling high-dimensional data and recognizing complex patterns through various kernel functions is advantageous. Furthermore, training data in this work employs SVM with a linear kernel due to its speed and simplicity.

Also, in SVM, C is a regularization parameter defining the misclassification cost, influencing the model's bias and variance. A higher C signifies low bias and high variance, while a lower C implies high bias and low variance. The value of 1.0 is chosen for C in this study after experimentation with various values, including 0.1, 1.0, 10, and 100, using trial and error. This choice strikes a balance, ensuring the model avoids both overfitting and underfitting, contributing to its optimal performance. The model's exceptional accuracy justifies the superior performance of Linear SVM. DT+InceptionV3, NB+MobileNetV3, and KNN+ResNet152V2 demonstrate the lowest performance with accuracies of 43.31%, 44.49%, and 60.16%, respectively. Particularly noteworthy is the minimum accuracy of Linear SVM in combination with MobileNetV2, which stands at 79.12%, signifying a substantial 35.81% improvement compared to DT+InceptionV3. This difference indicates that the proposed model significantly outperforms the lowest-performing model overall.

The findings unveil a compelling trade-off between model complexity and accuracy, where the proposed DenseNet201+Linear SVM model emerges as the optimal choice, outperforming other combinations. Notably, all other DL+ML model combinations exhibit lower accuracy than the suggested model, coupled with a more significant number of parameters and sizes (refer to Table 3). These results underscore the unique synergy between DenseNet201's feature extraction capabilities and the discriminative power of SVM in achieving state-of-the-art accuracy in multi-crop leaf disease identification. This capability enables the model to be lightweight and detect diseases even with limited training datasets. Also, the outcomes validate the practicality of the suggested model in current real-time farming settings.

6 Conclusion

The prevalence of crop diseases significantly threatens global food security, underscoring the importance of early detection and preventive measures. Corn, Rice, and Wheat are widely consumed crops worldwide, and prompt detection of diseases impacting them is crucial to ensure the sustainability of the food supply. Despite numerous Machine Learning and Deep Learning models available for disease classification, practical implementation needs improvement in an environment with limited resources. The research presents a hybrid model designed for identifying diseases in Corn, Rice, and Wheat crops, aiming to tackle these challenges effectively. The proposed approach streamlines disease classification by offering a singular model, minimizing the need for crop-specific models. Moreover, it provides a practical solution for environments with limited resources, utilizing only 20.2M parameters. Significantly, the model attains an accuracy of 87.23%, outperforming various benchmark models assessed in this study.

Extensive experimentation demonstrates the effectiveness of the proposed model across diverse scenarios, specifically in the individual disease classification of Corn, Rice, and Wheat. However, the model training and performance are limited to three cereal crops. Hence, training on a broad spectrum of datasets is required to improve the model's generalizability for a more comprehensive range of crops. Further, ML model monitoring tools such as TensorBoard, MLflow, or Prometheus can be effectively used in the production environment to streamline monitoring and ensure stability, accuracy, and efficiency. The hybrid and lightweight nature of the proposed model opens up possibilities for integration with handheld devices in the future, enabling farmers to detect diseases easily. Additionally, there is potential for implementing the model on drones, thus alleviating the necessity for labor-intensive manual inspection. Furthermore, disease classification can be enhanced by incorporating in-situ data from crop fields alongside the images, allowing for more accurate disease identification.

Data Availibility The dataset used is publicly available as cited in the article.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

References

1. Nations U (2022) World's Population. https://www.un.org/en/global-issues/population. Accessed 31 July 2023
2. of Agriculture & Farmers Welfare M (2023) Contribution of agricultural sector in GDP. https://www.pib.gov.in/PressReleasePage.aspx?PRID=1909213. Accessed 31 July 2023
3. Wire T (2023) Share of agriculture in employment rose, manufacturing declined In 2021-22: PLFS. https://thewire.in/economy/share-of-agriculture-in-employment-rose-manufacturing-declined-in-2021-22-plfs. Accessed 31 July 2023
4. Today CA (2022) Pests and diseases cause worldwide damage to crops. https://californiaagtoday.com/pests-diseases-cause-worldwide-damage-crops/. Accessed 31 July 2023
5. Garg G, Gupta S, Mishra P, Vidyarthi A, Singh A, Ali A (2023) CROPCARE: an intelligent real-time sustainable IoT system for crop disease detection using mobile vision. IEEE Internet of Things J 10(4):2840–2851
6. Sinha A, Shrivastava G, Kumar P (2019) Architecting user-centric internet of things for smart agriculture. Sustain Comput Inform Syst 23:88–102
7. Srinivas L, Bharathy AV, Ramakuri SK, Sethy A, Kumar R (2023) An optimized machine learning framework for crop disease detection. Multimed Tools Appl 1–20
8. Bhola A, Kumar P (2023) Performance evaluation of different machine learning models in crop selection. In: Robotics, control and computer vision: select proceedings of ICRCCV 2022 pp 207–217. Springer
9. Liu K, Zhang X (2022) PiTLiD: identification of plant disease from leaf images based on convolutional neural network. IEEE/ACM Trans Comput Biol Bioinform 20(2):1278–1288
10. Bhola A, Verma S, Kumar P (2023) A comparative analysis of deep learning models for cucumber disease classification using transfer learning. J Current Sci Technol 13(1):23–35
11. Tian L, Zhang H, Liu B, Zhang J, Duan N, Yuan A, Huo Y (2023) VMF-SSD: a novel v-space based multi-scale feature fusion SSD for apple leaf disease detection. IEEE/ACM Trans Comput Biol Bioinform 20(3):2016–2028
12. Food of the United Nations AO, Dimensions of need - Staple foods: what do people eat? https://www.fao.org/3/u8480e/u8480e07.htm. Accessed 10 July 2023
13. Khan S, Narvekar M (2022) Novel fusion of color balancing and superpixel based approach for detection of tomato plant diseases in natural complex environment. J King Saud University-Comput Inf Sci 34(6):3506–3516
14. Kaur S, Pandey S, Goel S (2018) Semi-automatic leaf disease detection and classification system for soybean culture. IET Image Proc 12(6):1038–1048

15. Basavaiah J, Arlene Anthony A (2020) Tomato leaf disease classification using multiple feature extraction techniques. Wirel Pers Commun 115(1):633–651
16. Kalyoncu C, Toygar Ö (2016) GTCLC: leaf classification method using multiple descriptors. IET Comput Vision 10(7):700–708
17. Zamani AS, Anand L, Rane KP, Prabhu P, Buttar AM, Pallathadka H, Raghuvanshi A, Dugbakie BN (2022) Performance of machine learning and image processing in plant leaf disease detection. J Food Qual 2022:1–7
18. Rahman SU, Alam F, Ahmad N, Arshad S (2023) Image processing based system for the detection, identification and treatment of tomato leaf diseases. Multimed Tool Appl 82(6):9431–9445
19. Ahmed I, Yadav PK (2023) Plant disease detection using machine learning approaches. Expert Syst 40(5):13136
20. Prabu M, Chelliah BJ (2023) An intelligent approach using boosted support vector machine based arithmetic optimization algorithm for accurate detection of plant leaf disease. Pattern Anal Appl 26(1):367–379
21. Chen Y, Chen X, Lin J, Pan R, Cao T, Cai J, Yu D, Cernava T, Zhang X (2022) DFCANet: a novel lightweight convolutional neural network model for corn disease identification. Agriculture 12(12):2047
22. Genaev M, Ekaterina S, Afonnikov D (2020) Application of neural networks to image recognition of wheat rust diseases. In: 2020 cognitive sciences, genomics and bioinformatics (CSGB). IEEE, pp 40–42
23. Faisal M, Leu J-S, Avian C, Prakosa SW, Köppen M (2023) DFNet: dense fusion convolution neural network for plant leaf disease classification. Agron J
24. Chen J, Chen W, Zeb A, Yang S, Zhang D (2022) Lightweight inception networks for the recognition and detection of rice plant diseases. IEEE Sensors J 22(14):14628–14638
25. Wang D, Wang J, Li W, Guan P (2021) T-CNN: trilinear convolutional neural networks model for visual detection of plant diseases. Comput Electron Agric 190:106468
26. Zhu X, Li J, Jia R, Liu B, Yao Z, Yuan A, Huo Y, Zhang H (2022) LAD-Net: a novel light weight model for early apple leaf pests and diseases classification. IEEE/ACM Trans Comput Biol Bioinform 20(2):1156–1169
27. Ganatra N, Patel A (2020) Performance analysis of fine-tuned convolutional neural network models for plant disease classification. Int J Autom Control 13(3):293–305
28. Paymode AS, Malode VB (2022) Transfer learning for multi-crop leaf disease image classification using convolutional neural network VGG. Artif Intell Agric 6:23–33
29. Zhou C, Zhou S, Xing J, Song J (2021) Tomato leaf disease identification by restructured deep residual dense network. IEEE Access 9:28822–28831
30. Verma S, Kumar P, Singh JP (2023) A unified lightweight CNN-based model for disease detection and identification in Corn, Rice, and Wheat. IETE J Res 1–12
31. Haridasan A, Thomas J, Raj ED (2023) Deep learning system for paddy plant disease detection and classification. Environ Monit Assess 195(1):120
32. Aishwarya N, Praveena N, Priyanka S, Pramod J (2023) Smart farming for detection and identification of tomato plant diseases using light weight deep neural network. Multimed Tool Appl 82(12):18799–18810
33. Vieira G, GitHub - gabrieldgf4/PlantVillage-Dataset. https://github.com/gabrieldgf4/PlantVillage-Dataset. Accessed 10 July 2023
34. Singh D, Jain N, Jain P, Kayal P, Kumawat S, Batra N (2020) PlantDoc: a dataset for visual plant disease detection. In: Proceedings of the 7th ACM IKDD CoDS and 25th COMAD. pp 249–253
35. Saleem MH, Potgieter J, Arif KM (2020) Plant disease classification: a comparative evaluation of convolutional neural networks and deep learning optimizers. Plants 9(10):1319
36. Singh UP, Chouhan SS, Jain S, Jain S (2019) Multilayer convolution neural network for the classification of mango leaves infected by anthracnose disease. IEEE access 7:43721–43729
37. Akkem Y, Biswas SK, Varanasi A (2023) Smart farming using artificial intelligence: a review. Eng Appl Artif Intell 120:105899
38. Orchi H, Sadik M, Khaldoun M, Sabir E (2023) Automation of crop disease detection through conventional machine learning and deep transfer learning approaches. Agriculture 13(2):352
39. Akkem Y, Biswas SK, Varanasi A (2023) Smart farming monitoring using ML and MLOps. In: International conference on innovative computing and communication. Springer, pp 665–675
40. RIYAZ S, Rice Leafs. https://www.kaggle.com/datasets/shayanriyaz/riceleafs. Accessed 10 July 2023
41. GETCH O, Wheat Leaf dataset. https://www.kaggle.com/datasets/olyadgetch/wheat-leaf-dataset. Accessed 10 July 2023
42. HUSSAIN S , CIGAR computer vision for crop disease. https://www.kaggle.com/datasets/shadabhussain/cgiar-computer-vision-for-crop-disease. Accessed 10 July 2023

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.